Azure Log Analytics (LAW) vs. Splunk

We already have a Splunk license in the company, so why do we need Azure Log Analytics Workspace (LAW)? Can't we send everything to Splunk directly?
The Core Rule: Don't send everything everywhere. Send application errors and live traces to **LAW** so developers can fix issues quickly. Send security, firewall, and audit logs to **Splunk** for our enterprise security and compliance team. If you duplicate all logs to both tools, your monthly bill will become huge.

1. What Problems Are We Solving With This Strategy?

1. Slow Outage Recovery Time

Problem: Digging through raw text logs in Splunk during a production API 500 error takes 30+ minutes.
Solution: App Insights + LAW gives visual end-to-end trace maps, locating the exact SQL/Service Bus failure in 2-3 minutes.

2. Explosive Log Budget Inflation

Problem: Sending high-volume raw debug logs and metrics into Splunk burns license quotas quickly (~$1,800/mo).
Solution: Smart tiering in LAW (Basic/Analytics) cuts app debug costs by 80% (~$400/mo dual-layer).

3. Team Conflict & Tool Friction

Problem: Security teams don't want app debug noise in Splunk; Developers don't get visual request flow maps in Splunk.
Solution: Clear split—DevOps uses LAW for app health; Security Team uses Splunk for enterprise threats.

4. Infrastructure Maintenance Overhead

Problem: Maintaining VM log forwarders (Logstash/Launchpad) creates extra operational work.
Solution: 100% serverless pipeline using Azure Diagnostic Settings + Event Hubs with zero VM patching.

2. What These Technical Terms Mean (In Plain English)

Before jumping into tools, here is what these everyday telemetry and team terms actually mean:

- **Security & Compliance Team (SOC):** The central security team that monitors company-wide cyber threats, unauthorized logins, and firewall attacks across all cloud platforms (Azure, AWS, On-Prem).
- **Metrics (Numbers & Counters):** Simple numbers like CPU %, Memory %, or how many requests hit our app per minute. It is light, cheap to store, and shows *if system is slow or overloaded right now*.
- **Logs (Text Statements):** Plain text lines written by apps or Azure services when something happens. Example: `"Database connection timeout at 10:05 AM"`. It tells *why something broke*.
- **Traces (The Request Journey):** When a user clicks "Submit", that request travels: Web App → Service Bus → Database. Trace shows this full journey line-by-line so we easily find *which exact service took 10 seconds*.
- **Exceptions (Code Crashes):** Unhandled code failures (like null pointer or missing key). App Insights automatically captures exact file name and line number where code crashed.
- **Dependencies (External Calls):** Anything our code calls outside itself—like SQL DB, third-party API, or Service Bus Queue. App Insights measures how much time that external call wasted.

3. Azure Resources We Are Monitoring

Azure Resource	What Telemetry We Collect	Why We Need This Data
Azure Web Apps	HTTP 4xx/5xx status, request duration, stdout/stderr streams.	Find failing APIs and slow web requests quickly.
Azure Function Apps	Invocation counts, execution time, unhandled code exceptions.	Catch background job crashes and execution timeouts.
Azure Service Bus	Active messages, Dead-letter count, throttled request logs.	Stop queue bottlenecks and lost message processing.
Azure SQL & Postgres	CPU %, connection drops, deadlocks, slow query statements.	Fix database query bottlenecks and lockup issues.

Azure Resource	What Telemetry We Collect	Why We Need This Data
Azure Cosmos DB	Request Units (RU/s) consumption, HTTP 429 throttling events.	Check capacity limits and stop database rate limiting.
Storage Account (Blob)	Read/Write transaction failures, HTTP 403 access blocks.	Troubleshoot broken file uploads and permission errors.
Application Gateway	Backend health status, 502/504 errors, WAF rule matches.	Monitor gateway routing and incoming firewall attacks.

4. How We Route Data (No Extra VM Maintenance)

We use pure native Azure capabilities to route logs so we don't need to maintain separate forwarder VMs (like Logstash):

- **Application Insights (Automatic):** Attached to Web Apps & Function Apps. Automatically streams request traces, dependencies, and exceptions to LAW without manual code changes.
- **Azure Diagnostic Settings:** Configured on Service Bus, SQL, Cosmos DB, and Storage to push resource-level diagnostic logs into LAW.
- **Azure Event Hubs (To Splunk HEC):** Diagnostic Settings send Application Gateway WAF blocks and audit logs into Event Hub, which streams directly to Splunk HTTP Event Collector.

5. LAW vs. Splunk Comparison (Pros & Cons)

Tool	Pros (Why it is good)	Cons (Where it struggles)
Azure LAW + App Insights	• **DevOps Troubleshooting:** Instantly pinpoints app 500 errors and slow SQL queries in 2 minutes. • **Zero Setup:** Web Apps and Function Apps send traces automatically. • **Cheaper App Storage:** Basic table tiers cut debug log costs by ~80%. • **Included Dashboards:** Uses Azure Workbooks inside the portal without extra UI licenses.	• **Azure Focus Only:** Not ideal for monitoring non-Azure servers or AWS in one central place. • **Not an Enterprise SIEM:** Not designed for full security threat monitoring across the entire company.
Splunk Enterprise	• **Single Security View:** Combines Azure, AWS, network firewalls, and on-prem logs in one central dashboard. • **Built for Security Teams:** Excellent for security analysts tracking user logins, attacks, and company compliance. • **Powerful Log Search:** Very flexible query language to analyze security patterns across systems.	• **Expensive for Raw Debug Logs:** Streaming raw application debug traces burns Splunk license budget very fast. • **No Visual App Maps:** Does not give instant visual request maps like App Insights out-of-the-box.

6. LAW Table Plans (Controlling Log Costs)

We use three table tiers in LAW to avoid paying full ingestion price for all logs:

Table Tier	Cost Level	How We Use It
Analytics Tier	Standard (~$2.30/GB)	Use for active application errors, 5xx status codes, and live alert rules. Full KQL search.
Basic Tier	~80% Cheaper (~$0.50/GB)	Use for high-volume debug logs needed only for occasional troubleshooting. Pay per search.
Auxiliary Tier	~90% Cheaper (~$0.05/GB)	Use for high-volume raw compliance logs kept long-term. Searched via background jobs.

7. Data Routing Rules (Avoiding Log Duplication)

Golden Rule: Never send identical raw log lines to both LAW and Splunk. Route data according to who actually reads it.

Send to LAW (For Developers & DevOps)	Send to Splunk (For Security & Compliance Team)
• Application Insights Traces, Exceptions & Dependencies • Web App & Function App console log streams • Service Bus Dead-Letter errors & queue metrics • Azure SQL & Postgres slow queries & connection failures • Cosmos DB 429 rate limit events	• App Gateway WAF Blocked Attack Logs (`AGWWAFFirewallLogs`) • Entra ID (Azure AD) User Sign-ins & Audit Logs • Key Vault Secret Access Audit Trails • Cross-Cloud (Azure + AWS + On-Prem) Security Logs • Long-term Regulatory Compliance Archives

8. Estimated Cost Comparison (10 GB / Day Workload)

Approach	Est. Monthly Spend	Practical Result
Smart Split Strategy (Operational in LAW, Security in Splunk)	**~$380 – $420 / month**	**BEST CHOICE:** Fast developer troubleshooting in LAW + full security in Splunk with optimal cost.
Stream Everything to Splunk Only	**~$1,000 – $1,200+ / month**	Consumes Splunk quota quickly; hard to trace Azure app bugs visually.
Duplicate Everything (100% logs to LAW AND Splunk)	**~$1,800+ / month**	**WORST CHOICE:** Paying twice for the exact same raw data rows.

9. How Dashboards & Alerts Work in Practice

Instead of mixing everything into one place, we use each tool where its dashboards and alerts make the most sense for our daily work:

Capability	Azure LAW + Application Insights	Splunk Enterprise
Dashboards	• **Azure Workbooks:** Built right into the Azure Portal with no extra license fee. • **What it shows:** Visual request flow maps, Live Metrics streams, SQL query latency spikes, and server CPU/Memory health. • **Daily Use:** Developers and DevOps use this during daily standups and live bug fixing.	• **Splunk Dashboards:** Main security dashboard for the central team. • **What it shows:** Firewall attack trends, failed login spikes, and compliance across Azure, AWS, and on-premises servers. • **Daily Use:** Security analysts and leadership use this for overall company threat visibility.
Alerts	• **Azure Monitor Alerts:** Fast alerts based on native Azure metrics and KQL log queries. • **Cost & Speed:** Native Azure metric alerts are virtually free and trigger within seconds. • **Daily Use:** Immediate alerts for HTTP 500 spikes, Service Bus dead-letter buildup, database connection drops, and code crashes via Teams, Email, or PagerDuty.	• **Splunk Search Alerts:** Alerts built using SPL search queries on security log streams. • **Cost & Speed:** Best for pattern matching across systems, but avoid running high-frequency alerts on raw app debug logs to save compute quota. • **Daily Use:** Alerts for malicious security threats (e.g., multiple failed logins followed by a WAF block).

Use Azure Workbooks & Azure Monitor Alerts for day-to-day app performance and outage fixes because it is instant and cheap. Use Splunk Dashboards & Search Alerts for company-wide security, firewall blocks, and compliance monitoring.